

Marcus Neo Jing Quan

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Founding ML Engineer with 4+ years at an AI startup, owning the full machine learning lifecycle — from training pipeline architecture and model export to inference serving and data annotation tooling. Built and scaled production systems across classical computer vision, 3D medical imaging, and vision-language model fine-tuning on GCP.

EDUCATION

Imperial College London

Oct 2018 – Jun 2022

MEng Electronic & Information Engineering, First Class Honours

- Final Year Project: *Data Stream Evolution using Multiresolution Wavelet Transform* — Python-based ML research applying wavelet transforms as preprocessing for anomaly detection in data streams

Hwa Chong Institution, Singapore

Jan 2014 – Dec 2015

GCE A-Levels — AAAA

EXPERIENCE

Datature

Jul 2022 – Present

Founding Machine Learning Engineer (Team Lead)

- Led a team of 3 ML engineers across training, inference, and annotation services

Model Training Service — Sole author, deployed on Google Compute Engine

- Enabled customers to train on any major framework without migration overhead by building a framework-agnostic pipeline supporting JAX, TensorFlow, and PyTorch with multi-GPU support (Data/Model Parallelism); 3,800+ models trained to date across manufacturing, medical, agriculture, sports, and military sectors
- Onboarded 16+ model architectures (YOLOv8/11/26, MaskRCNN, EfficientDet, DeepLabV3, PaliGemma, DFINE, RF-DETR, EfficientAD), giving users broad coverage across classification, detection, segmentation, pose estimation, anomaly detection, and 3D medical imaging
- Improved model performance on high-resolution imagery via sliding window (SAHI) training with NMS and configurable overlap ratios
- Reduced manual data preparation through class auto-balancing, multilabel classification (Brier score, co-occurrence confusion matrices), and custom training metadata integration
- Enabled faster iteration via continual learning, allowing users to resume from prior checkpoints without retraining from scratch
- Lowered the barrier for non-ML users by building an OpenAI-powered hyperparameter recommender for automated training configuration

3D Medical Imaging Pipeline — Built end-to-end using MONAI

- Unlocked a new product vertical in medical imaging by developing [end-to-end pipelines](#) for volumetric models (SwinUNETR, SegResNet, nnUNet, SegMamba) with NifTI loading, 3D augmentations, and affine-aware evaluation
- Streamlined data exchange with existing medical workflows via NifTI annotation import/export with class-indexed and one-hot mask exports, 3D TFRecord export, and aggregation statistics
- Extended clinical radiotherapy support by adding DICOM input and RTSTRUCT output
- Defined schemas and chunk encoding for 3D bitmask serialisation in the shared core type library used across all services
- Deployed into active clinical workflows, serving enterprise medical customers in clinical radiology and radiotherapy

Vision-Language Model Fine-Tuning Pipeline — Built from scratch, deployed on GCP

- Launched the VLM product line by building a dedicated fine-tuning service with full CI/CD, supporting [Qwen2.5-VL](#), [Qwen3-VL](#), NVILA, CosmosReason1/2, and Gemma4
- Reduced GPU memory requirements via LoRA fine-tuning with quantization; scaled multi-GPU training through tensor parallelism with OOM-resilient training loops
- Covered diverse customer use cases — VQA, Chain-of-Thought reasoning with structured evaluation, freeform generation, and video training via PyAV ingestion
- Automated annotation labelling at scale by building the Intelliscribe caption generation microservice with JSON schema validation for structured annotation outputs
- Extending the pipeline with tool-calling fine-tuning support and RLHF capabilities, enabling models to learn structured function invocation and preference-based alignment

Model Export Service — Authored, deployed on Google Cloud Run

- Enabled deployment to [edge devices](#), mobile, and real-time applications via cross-framework export (TFLite, CoreML, TensorRT, OpenVINO, ONNX) with [Float16/Int8 quantization](#) (up to 75% model size reduction) and [model pruning](#) up to 90% of parameters; 2,700+ model artifacts exported to date
- Hardened service reliability under heavy load by wrapping export conversion in multiprocessing with async apply and timeout to prevent Cloud Run request hangs

Model Hosting & Inference Service — Co-authored, deployed on Google Kubernetes Engine

- Powered production inference for all onboarded model architectures across image, video, and 3D volumetric inputs (NIfTI and DICOM) by integrating Triton Inference Server with Numba JIT optimisation
- Reduced annotation costs for customers by building an active learning pipeline with entropy-based metrics and bitmask annotation re-upload to the platform
- Shipping RAG and tool-calling capabilities into the VLM inference serving layer, enabling retrieval-augmented generation and structured function dispatch for production VLM deployments

Annotation Data Services

- Eliminated data migration friction via annotation import/export for 10+ formats (COCO, LabelMe, SuperAnnotate, ScaleAI, OpenAI, VoTT, Supervisely, IBM) with multipolygon, sliding window, and 3D NIfTI support; 3,800+ training datasets and 1,400+ standalone annotation exports to date
- Built a VQA/Visual Intelligence annotation pipeline: JSONL and ViFull import/export, phrase grounding with caption/bbox support, and Neural Insights
- Improved dataset quality assurance by implementing Mosaic and CutMix augmentation preview for training data visualisation

Video Annotation Services — Deployed on GKE with KEDA scale-from-zero

- Enabled frame-by-frame video annotation at scale by building video interpolation and SAM2-based tracking services with PubSub queue and multiprocessing to prevent GPU memory leaks
- Hardened production readiness with KEDA authentication, production configs, and memory-optimised caching

CI/CD & DevOps

- Accelerated release cycles by designing GitHub Actions CI/CD pipelines across all ML services — automated linting, testing, Docker builds, and deployment to GCP (Compute Engine, Cloud Run, GKE)
- Reduced deployment image sizes through multi-stage Docker builds, while managing CUDA runtime upgrades and Python version migrations across training, export, and inference services
- Owned and maintained PyPI and npm packages for the shared Alchemy training library and nexus-core-schema type library, ensuring version consistency across services

Datature

Apr 2021 – Jan 2022

Machine Learning Intern

- Built custom on-the-fly augmentations for the TensorFlow training pipeline and microservices for augmentation preview, dataset statistics aggregation, and hyperparameter recommendations
- Developed the full backend for Portal (neural net visualisation webapp): REST API, multi-framework model compatibility (TensorFlow, PyTorch, DarkNet), video/image predictions, and bulk inference
- Built and deployed an ML model hosting service on GCP using Docker

CORE COMPETENCIES

- **Languages:** Python, C++
- **ML Frameworks:** PyTorch, TensorFlow, JAX, ONNX, MONAI, Ultralytics
- **ML Serving & Export:** Triton Inference Server, TensorRT, CoreML, TFLite, OpenVINO, Numba
- **VLM Fine-Tuning:** LoRA, Quantization, Tensor Parallelism, DeepSpeed, HuggingFace
- **Cloud & Infrastructure:** GCP (Compute Engine, Cloud Run, GKE, PubSub), Docker, Kubernetes, KEDA, CUDA
- **Languages (Spoken):** English (native), Mandarin (native)

ACTIVITIES & INTERESTS

- **Scuba Diving** — PADI Advanced Open Water + Nitrox certified, 50+ logged dives (Bali, Indonesia; Ko Tao, Thailand; Malapascua, Philippines; Osezaki, Japan; Tioman, Malaysia)
- **Snowboarding** — Picking it up one run at a time (Norikura Kogen, Japan; Yabuli, China)
- **Contract Bridge** — Serious hobbyist; 2/1 Game Forcing, 15-17 NT, Jacoby 2NT, Bergen Raises, RKCB 1430